

SUJONG LEE

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EDUCATION

Nanyang Technological University (NTU), Singapore *Aug 2026 – May 2030 (Expected)*
Nanyang President's Graduate Scholarship
Ph.D., College of Computing and Data Science
Supervised by: *Atsushi, Nitanda*

Nanyang Technological University (NTU), Singapore *Aug 2020 – May 2026*
B.Eng., Electrical and Electronics Engineering (Minor in Mathematics)
Highest Distinction

RESEARCH INTERESTS

Generative Models	Diffusion Models, Flow Models and their theoretical foundations
Neural Samplers	Diffusion-based samplers
AI4Science	Machine Learning for Materials Discovery

My research interests broadly lie in applied mathematics for machine learning. I am specifically interested in mathematical understanding and efficiency of neural network and generative model. Recently, I am studying diffusion model, flow model and their variations in connection to optimal transport and stochastic control.

PUBLICATIONS

1. **Sujong Lee**, J. Bae, S. Shin, H. Choi, C. Park. “Density of States-Intermediated Crystal Generation for Material Inverse Design.” *AI4Science@ICML, 2026*
2. **Sujong Lee**, P. Jutras-Dubé, B. Wen, R. Zhang. “Neural Discrete Controlled Monte Carlo Samplers.” *ProbML, 2026*

RESEARCH EXPERIENCES

Undergraduate Research Intern *Jun 2025 - Present*
Supervised by: *Ruqi, Zhang* *Purdue University, Remote*

- Research Topic: Sampling problem in discrete domain
- Discrete diffusion model for simulation of the CTMC process with Gumbel-Softmax relaxation for computation of the terminal cost
- Controlled CTMC by neural network to construct a discrete analogue of controlled Langevin Monte Carlo method

Undergraduate Research Intern *Jan 2025 - May 2025*
Supervised by: *Wen, Bihan* *ROSE Lab@NTU, Singapore*

- Research Topic: Generative AI empowered synthetic data/image generation
- Accelerating inference speed in diffusion model via Hamilton-Jacobi regularization
- Imposing architectural constraint and auxiliary term for theoretical guarantees of straight trajectory in single training

Undergraduate Research Intern *Mar 2022 - Jun 2022*
Supervised by: *Taehyoung (Tony), Kim* *NTU, Singapore*

- Research Topic: Design and Analysis of Neural Network
- Investigation of basic CNN models such as LeNet and ResNet

Undergraduate Research Experience on Campus (URECA)

Supervised by: *Donguk, Nam*

Sep 2021 - Jun 2022

NTU, Singapore

- Research Topic: Strain-engineered quantum device towards integrated quantum photonic chips
- Comparative analysis on PLE spectrum of various semiconductors for photonic waveguide
- FDTD simulational analysis using Ansys Lumerical software

WORK EXPERIENCES

AI Researcher Intern

NanoforgeAI, Korea

Oct 2025 - Present

- Density of States generation by Masked Diffusion Model
- Mattergen fine-tuning for DoS-conditioned crystal generation as a pipeline
- Publication under preparation on crystal generation pipeline with novel conditioning method

AI Research Engineer Intern

Paradot (Carat), Korea

Mar 2024 - Jun 2024

- Launched a new service on diffusion-based appearance transformation
- Migrated existing T2I image generation service from external API to SDXL with optimization of parameter and inference

Sergeant

SEC Research Center, Korea

Sep 2022 - Mar 2024

- National service in Republic of Korea Army (2022-2024)
- Developed and modified an internal program for military use

HONOURS/AWARDS

Nanyang President's Graduate Scholarship

NTU President Research Scholar (UG)

Dean's List AY2024/2025

NTU's most prestigious Ph.D. scholarship

Awarded to the Top 5% of the Cohort

SKILLS/HOBBIES

Programming Languages

Python, C++

Machine Learning Tools

PyTorch, Pandas, Numpy, Diffusers, Wandb